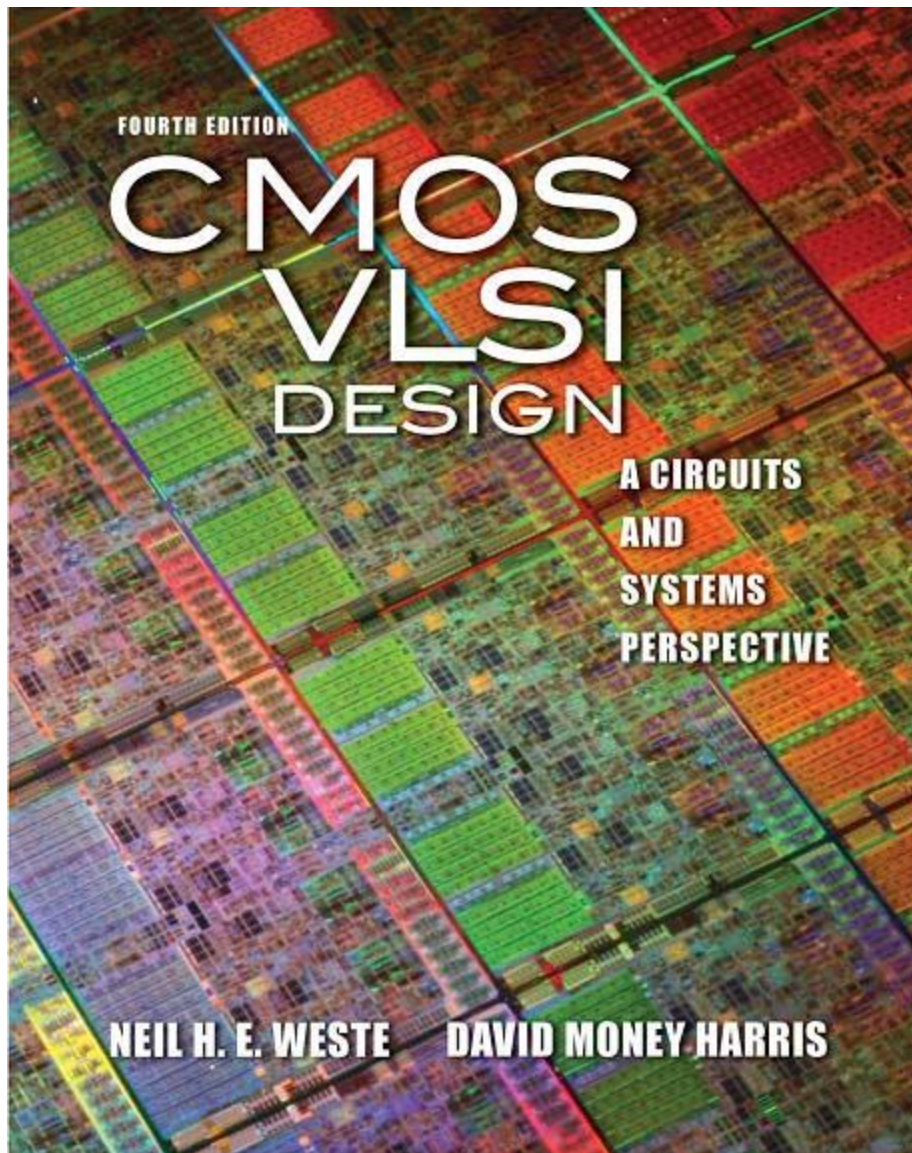


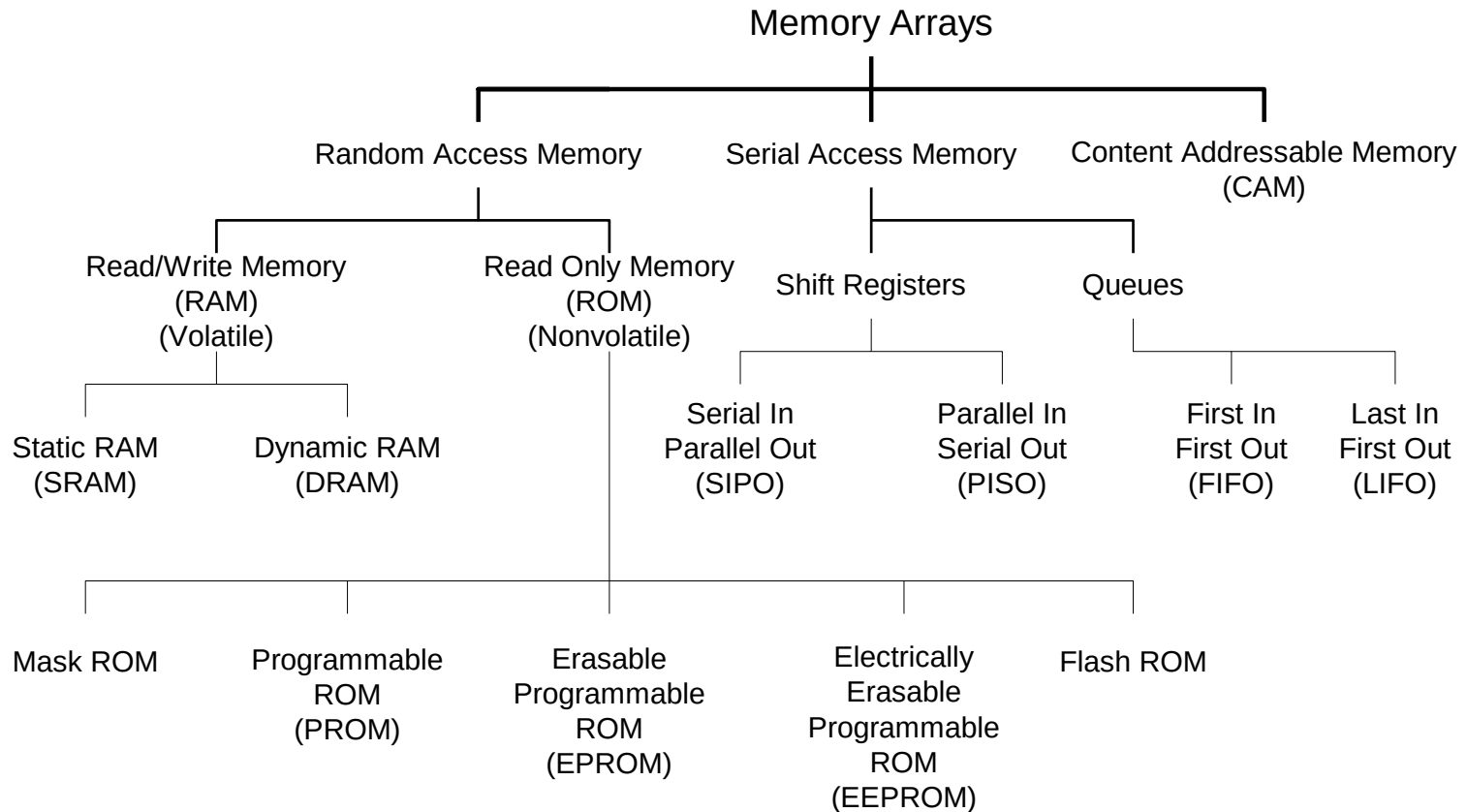


Array Subsystems

Lecture # 5

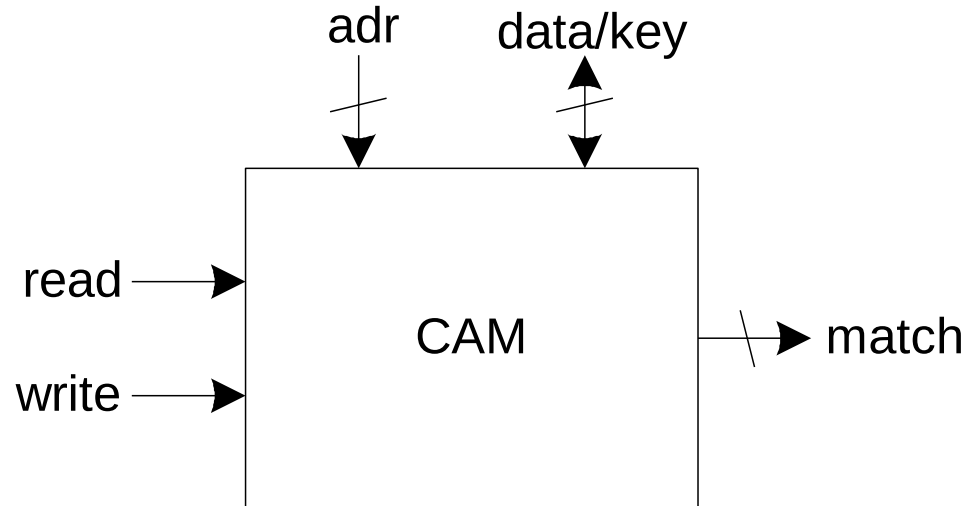
- ☐ Content-Addressable Memories
- ☐ CAM Cell
- ☐ CAM Cell Operation
- ☐ TLB Application

Memory Arrays



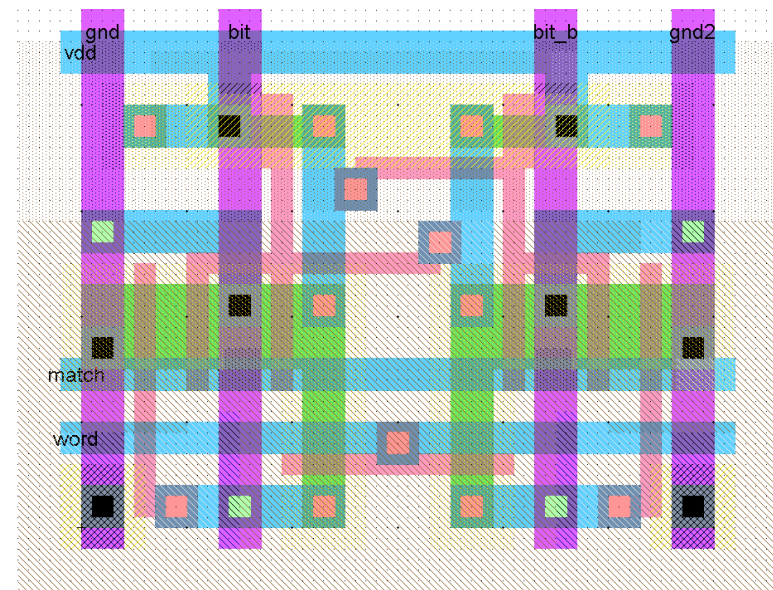
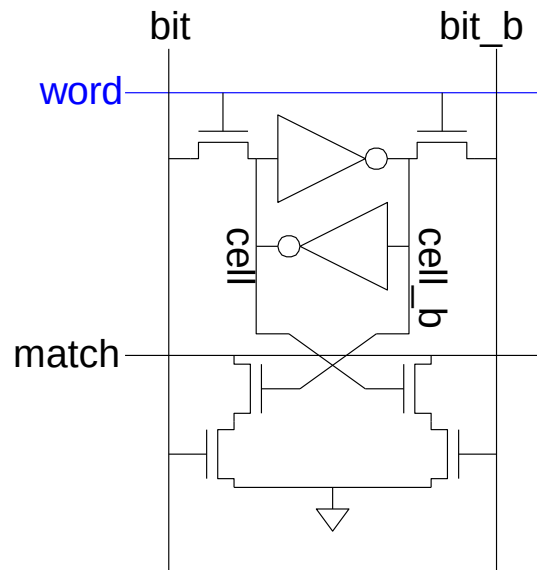
CAMs

- ❑ Extension of ordinary memory (e.g. SRAM)
 - Read and write memory as usual
 - Also *match* to see which words contain a *key*



10T CAM Cell

- ❑ Add four match transistors to 6T SRAM
 - 56 x 43 λ unit cell



CAM Cell Operation

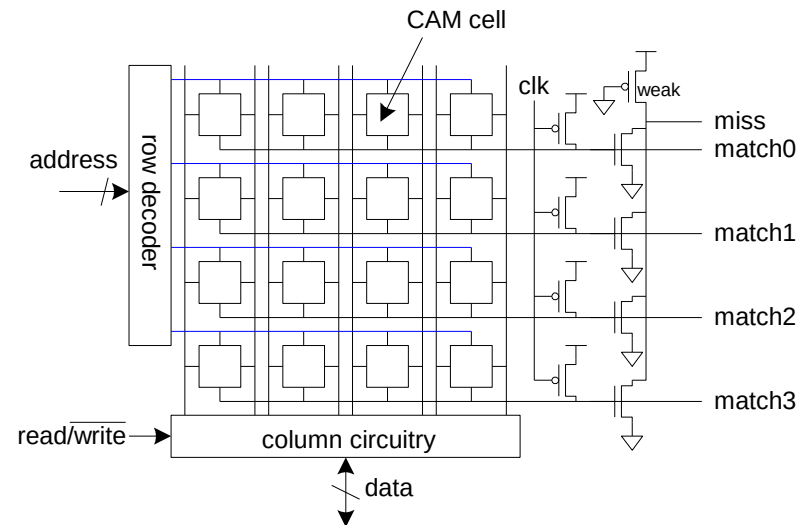
- ❑ Read and write like ordinary SRAM

- ❑ For matching:

- Leave wordline low
- Precharge matchlines
- Place key on bitlines
- Matchlines evaluate

- ❑ Miss line

- Pseudo-nMOS NOR of match lines
- Goes high if no words match



TLB

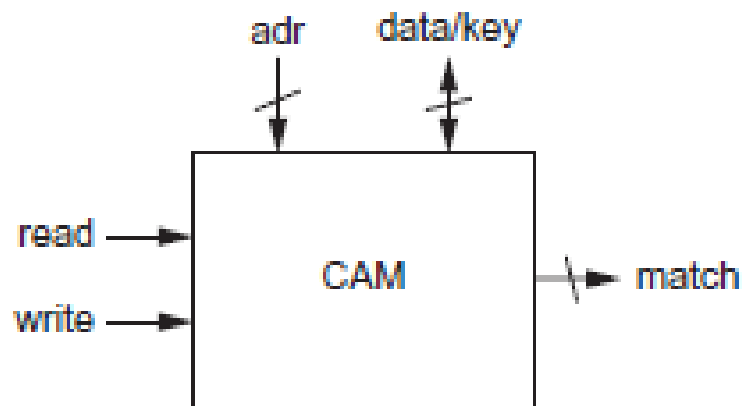


FIGURE 12.67 Content-addressable memory

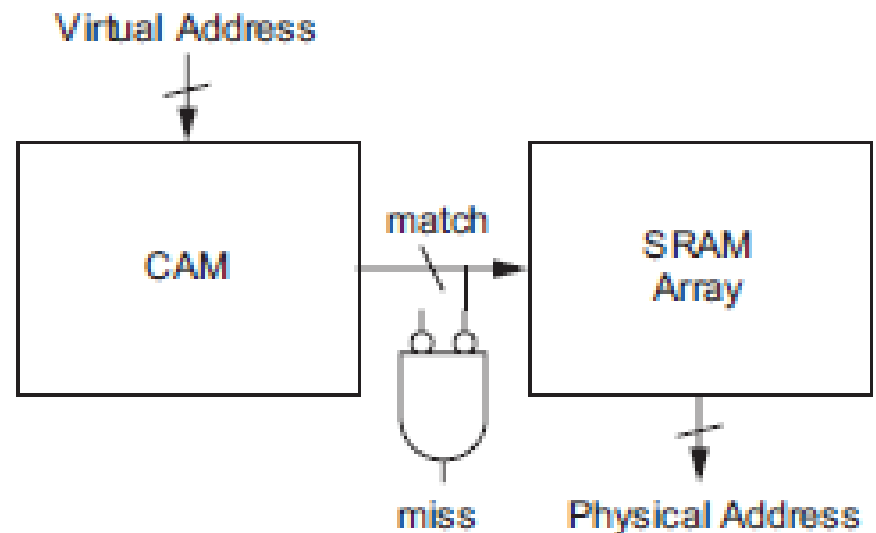


FIGURE 12.68 Translation Lookaside Buffer (TLB) using CAM

Monotonic Wordlines

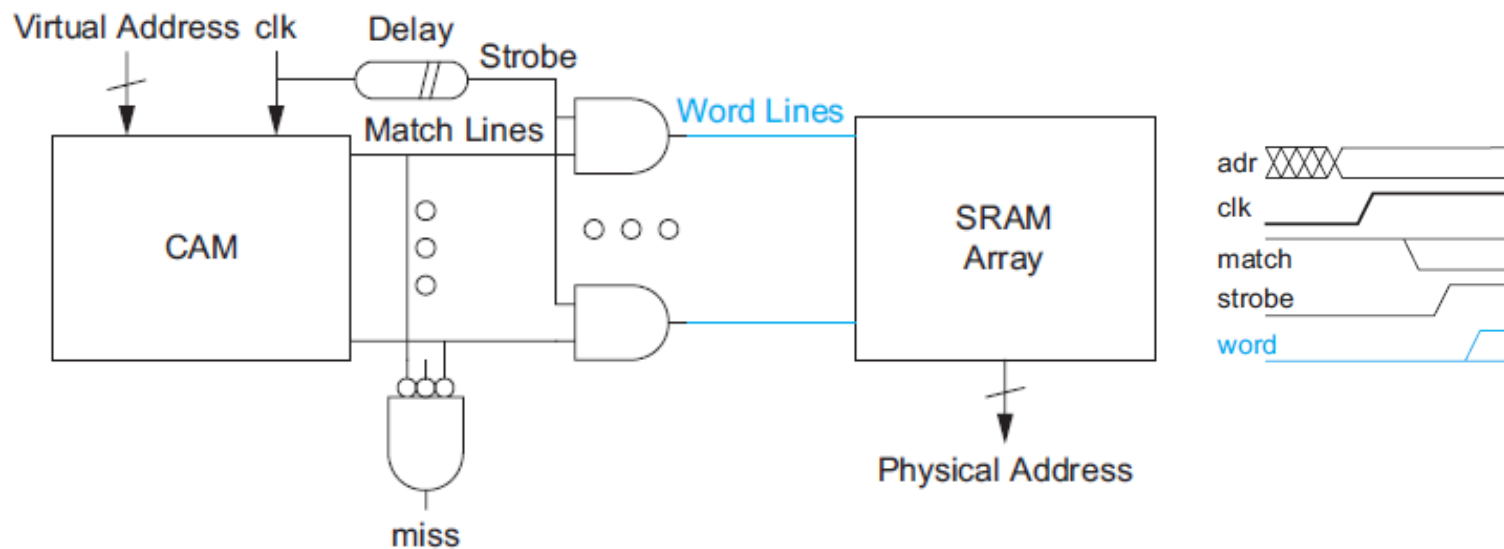


FIGURE 12.71 Refined TLB path with monotonic wordlines

Homework

- ❑ Design an 8 x 8 CAM array and identify the sizes of row decoders and column circuitry.